Fake News Detection Using Machine Learning (TF-IDF and Logistic Regression)

Problem Statement:

The rapid growth of social media and online news platforms has made it easy for false and misleading information to spread quickly. Detecting fake news manually is difficult and time-consuming because of the large amount of information available online, and it cannot keep pace with the volume of articles published every day.

There is, therefore, a need for an automated system that can analyze the textual content of a news article and classify it as Fake News or Real News. This project, "Fake News Detection Using Machine Learning", uses Natural Language Processing techniques to preprocess the news data and TF-IDF (Term Frequency-Inverse Document Frequency) to convert text into numerical features. These features are given to a Logistic Regression machine learning model for classification. The model is trained using a dataset containing both fake and real news articles and evaluated using Accuracy, Precision, Recall, F1-Score and a Confusion Matrix. The developed system can provide a quick prediction for a given news article and demonstrates the practical application of Machine Learning and Natural Language Processing in identifying misinformation.

Objectives:

1. To load and combine a labelled dataset of fake and real news articles (Fake.csv and True.csv).
2. To preprocess the data by labelling, removing unnecessary columns and missing values, and combining the news title and text into a single content field.
3. To convert the news text into numerical features using TF-IDF vectorization.
4. To train a Logistic Regression classifier on the extracted TF-IDF features.
5. To evaluate the trained model on an unseen test set and report Accuracy, Precision, Recall, F1-score and the Confusion Matrix.
6. To build a prediction function that classifies any given news article as FAKE NEWS or REAL NEWS.

Software / Hardware Required:

• Operating System: Windows / Linux / macOS
• Platform / IDE: Google Colab (Jupyter Notebook)
• Language: Python 3.x
• Machine Learning: scikit-learn (TfidfVectorizer, LogisticRegression)
• Data Handling: pandas, NumPy
• Visualization: Matplotlib
• Dataset: Fake and Real News dataset (Fake.csv – 23,481 articles, True.csv – 21,417 articles)

Project File Structure:

The project ("FakeNewsDetection") is organized as a single Google Colab notebook together with the two dataset files (Fake.csv and True.csv), as shown below. The notebook performs preprocessing, vectorization, training, evaluation and prediction.

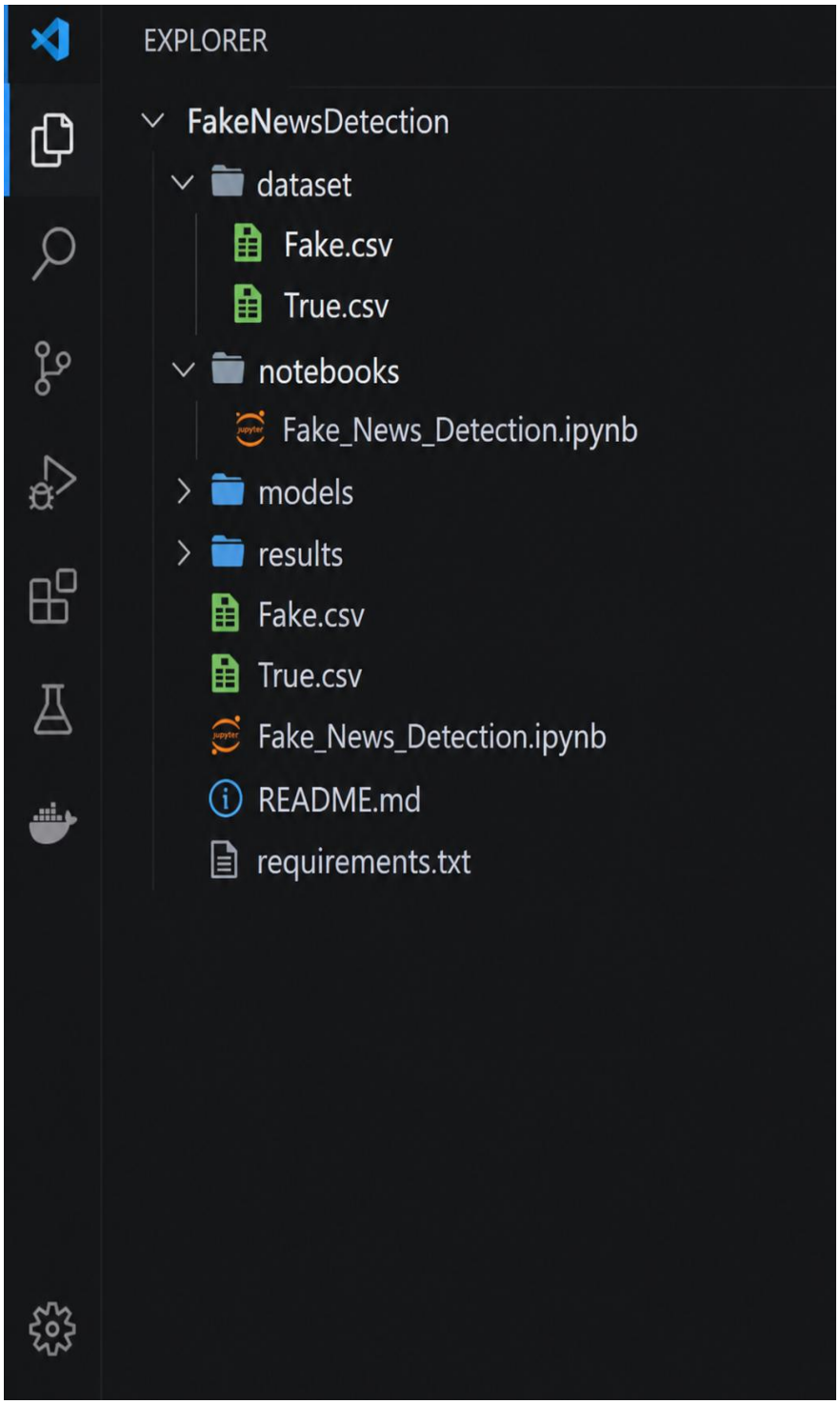


Fig 1: Project file structure of FakeNewsDetection in Visual Studio Code

Procedure:

1. Upload the dataset files (Fake.csv and True.csv) to Google Colab and load them using pandas.
2. Add labels to the data — 0 for Fake News and 1 for Real News — and combine both datasets into a single DataFrame.
3. Check the dataset information, keep only the required columns (title, text, label) and drop missing values.
4. Combine the news title and text into a single "content" column.
5. Check the count of fake and real news articles and visualize the class distribution using a bar chart.
6. Split the data into input (content) and output (label) and divide it into training (80%) and testing (20%) sets.
7. Convert the text into numerical features using TF-IDF vectorization (English stop words removed, max_df = 0.7, 50,000 features).
8. Train the Logistic Regression model on the TF-IDF features of the training data (max_iter = 1000).
9. Make predictions on the test data and calculate the Accuracy, Classification Report and Confusion Matrix.
10. Define a prediction function that takes a news article and returns FAKE NEWS or REAL NEWS, and test it on a sample article.

Implementation Code:

Implementation of the key steps of the project, as executed in the Google Colab notebook.

1. Import the Required Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
```

2. Upload and Load the Dataset

```
from google.colab import files
uploaded = files.upload()
```

```
fake = pd.read_csv("Fake.csv")
true = pd.read_csv("True.csv")
print("Fake News Dataset Shape:", fake.shape)
print("True News Dataset Shape:", true.shape)
```

3. Add Labels and Combine Both Datasets

```
fake["label"] = 0
true["label"] = 1

df = pd.concat([fake, true], axis=0)
print("Combined Dataset Shape:", df.shape)
df.head()
```

4. Check Dataset Information and Remove Unnecessary Columns

```
df = df[["title", "text", "label"]]
df = df.dropna()

df = df[["title", "text", "label"]]
df = df.dropna()
df.head()
```

5. Combine Title and News Text

```
df["content"] = df["title"] + " " + df["text"]
df = df[["content", "label"]]
df.head()
```

6. Check Fake and Real News Count and Visualize Dataset

```
print(df["label"].value_counts())

df["label"].value_counts().plot(
    kind="bar"
)
```

```
plt.xticks([0, 1], ["Fake News", "Real News"])
plt.xlabel("News Type")
plt.ylabel("Count")
plt.title("Fake vs Real News Distribution")

plt.show()
```

7. Split Input/Output and Train-Test Split

```
X = df["content"]
y = df["label"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
print("Training Data:", X_train.shape)
print("Testing Data:", X_test.shape)
```

8. TF-IDF Vectorization

```
vectorizer = TfidfVectorizer(
    stop_words="english",
    max_df=0.7,
    max_features=50000
)
X_train_vectorized = vectorizer.fit_transform(X_train)
X_test_vectorized = vectorizer.transform(X_test)
print("Vectorization Completed!")
```

9. Train Logistic Regression Model

```
model = LogisticRegression(
    max_iter=1000
)
model.fit(
    X_train_vectorized,
```

```
    y_train
)
print("Model Training Completed!")
```

10. Make Predictions and Calculate Accuracy

```
y_pred = model.predict(X_test_vectorized)
print(y_pred[:10])

accuracy = accuracy_score(y_test, y_pred)
print("Model Accuracy:", accuracy * 100, "%")
```

11. Classification Report and Confusion Matrix

```
print(classification_report(
    y_test,
    y_pred,
    target_names=["Fake News", "Real News"]
))

cm = confusion_matrix(y_test, y_pred)
print(cm)

plt.figure(figsize=(6, 5))
plt.imshow(cm)
plt.colorbar()
plt.xticks([0, 1], ["Fake", "Real"])
plt.yticks([0, 1], ["Fake", "Real"])
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```

12. Prediction Function for New News

```
def predict_news(news):
    news_vector = vectorizer.transform([news])
    prediction = model.predict(news_vector)
    if prediction[0] == 0:
```

```
        return "FAKE NEWS"
    else:
        return "REAL NEWS"

news = """Scientists confirm that a new technology can improve renewable energy
production."""
result = predict_news(news)
print("Prediction:", result)
```

Output (Step-by-Step Results):

```
Fake News Dataset Shape: (23481, 4)
True News Dataset Shape: (21417, 4)
```

Fig 2: Step 1 — Dataset loaded: shape of the Fake and True news datasets

Title	Text	Subject	Date	Label
Donald Trump Sends Out Embarrassing New Year...	Donald Trump just couldn't wish all Americans...	News	December 31, 2017	0
Drunk Bragging Trump Staffer Started Russian...	House Intelligence Committee Chairman Devin...	News	December 31, 2017	0
Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwauk...	News	December 30, 2017	0
Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announced that...	News	December 29, 2017	0
Pope Francis Just Called Out Donald Trump...	Pope Francis used his annual Christmas Day mes...	News	December 25, 2017	0

Fig 3: Step 2 — Combined dataset (44898 rows, 5 columns) after adding labels (0 = Fake, 1 = Real)

	title	text	label
0	Donald Trump Sends Out Embarrassing New Year'...	Donald Trump just couldn t wish all Americans ...	0
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Devin Nu...	0
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwauk...	0
3	Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announced that ...	0
4	Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day mes...	0

Fig 4: Step 3 — Dataset after removing unnecessary columns (title, text, label)

	content	label
0	Donald Trump Sends Out Embarrassing New Year'...	0
1	Drunk Bragging Trump Staffer Started Russian ...	0
2	Sheriff David Clarke Becomes An Internet Joke...	0
3	Trump Is So Obsessed He Even Has Obama's Name...	0
4	Pope Francis Just Called Out Donald Trump Dur...	0

Fig 5: Step 4 — Title and text combined into a single "content" column

```
label
0    23481
1    21417
Name: count, dtype: int64
```

Fig 6: Step 5 — Count of Fake News (23,481) and Real News (21,417) articles

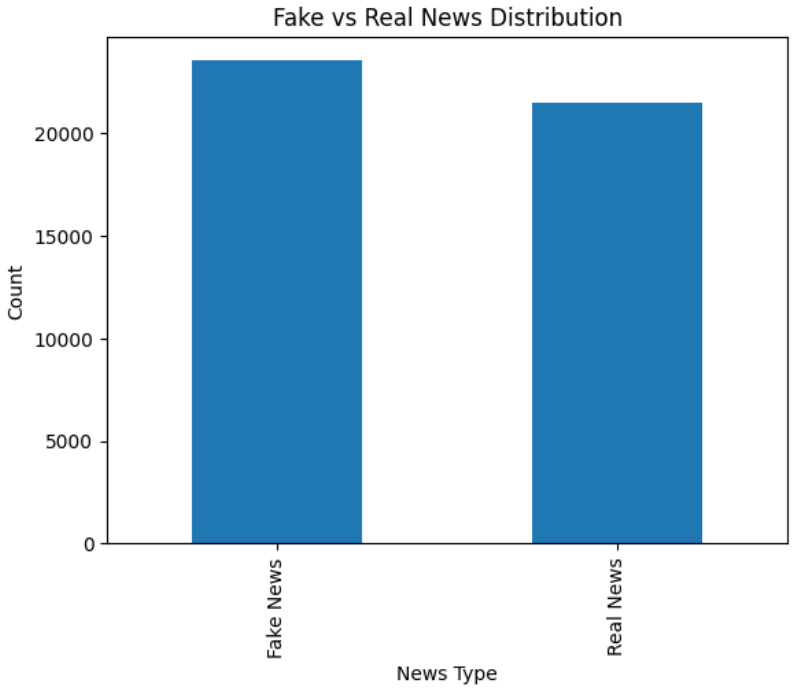


Fig 7: Step 5 — Fake vs Real News distribution bar chart

```
Training Data: (35918,)
Testing Data: (8980,)
```

Fig 8: Step 6 — Train-test split (80% training, 20% testing)

```
Vectorization Completed!
Model Training Completed!
[0 1 0 0 1 1 0 1 0 1]
```

Fig 9: Step 7 — TF-IDF vectorization, model training and first ten test predictions

```
Model Accuracy: 98.0 %
```

Fig 10: Step 8 — Model accuracy on the test set

	precision	recall	f1-score	support
Fake News	0.99	0.99	0.99	4733
Real News	0.98	0.99	0.99	4247
accuracy			0.99	8980
macro avg	0.99	0.99	0.99	8980
weighted avg	0.99	0.99	0.99	8980

Fig 11: Step 9 — Classification report (Precision, Recall, F1-Score) for Fake News and Real News

```
[[4663   70]
 [  55 4192]]
```

Fig 12: Step 10 — Confusion matrix values: [[TN, FP], [FN, TP]]

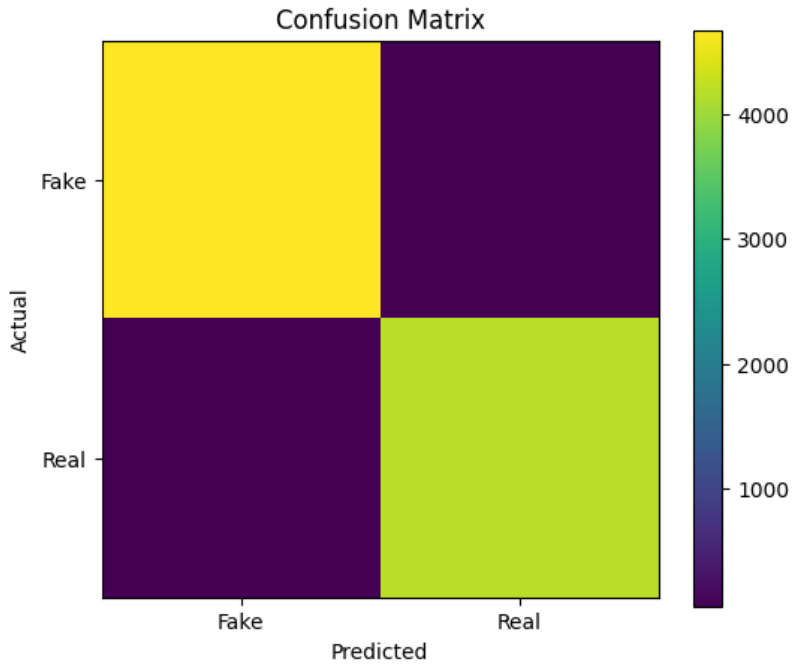


Fig 13: Step 10 — Confusion matrix plot (Actual vs Predicted: Fake / Real)

```
Prediction: real news
```

Fig 14: Step 11 — Prediction for a sample news article: "Scientists confirm that a new technology can improve renewable energy production."

Result:

A Fake News Detection system was successfully implemented using Natural Language Processing and Machine Learning. The news title and text were combined, converted into 50,000 TF-IDF features and classified by a Logistic Regression model trained on 35,918 articles. On the test set of 8,980 articles, the model achieved an accuracy of about 98%, with Precision, Recall and F1-Score of 0.99 for both Fake News and Real News, and only 125 misclassified articles in the confusion matrix (70 fake articles predicted as real and 55 real articles predicted as fake). The prediction function correctly classified a sample article as REAL NEWS.